

LESSON 4-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Find the Percent of a Number**

Maryland Standard 6.RP.A.3c • Learning target – I can find the percent of a number exactly.

REMEMBER

- The **whole** comes after “of.” In 30% of 80, the whole is 80.
- Percent to decimal, then multiply: $0.3 \times 80 = 24$.
- Or use 10%: 10% of 80 = 8, so $30\% = 3 \times 8 = 24$.

WORKED EXAMPLE

Find 35% of 60.

Step 1 Write 35% as a decimal: 0.35.**Step 2** Multiply: $0.35 \times 60 = 21$.**Step 3** Check with 10%: $6 + 6 + 6 + 3 = 21$.**Answer 21****WORD BANK****percent** – out of 100**whole** – the full amount; it comes after “of”**part** – the piece the percent tells about**decimal** – $45\% = 0.45$ **PRACTICE***show your work • label units*

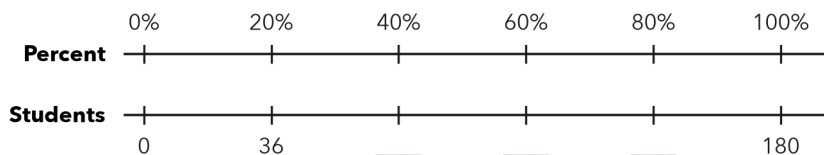
- 1.**
- Find each part.

Step 1 Write the percent as a decimal.**Step 2** Multiply the decimal by the number after “of.”

a. 20% of 45 = _____ **b.** 75% of 64 = _____ **c.** 6% of 350 = _____

d. 45% of 80 = _____ **e.** 150% of 30 = _____ **f.** 12% of 250 = _____

- 2. Lunch from Home.**
- A grade has 180 students. 40% of them bring lunch from home. Fill in the double number line. Then find how many students bring lunch.

Step 1 $180 \div 5 = 36$, so each jump is 20%.**Step 2** Add 36 each time to fill the line.**Step 3** Find the number under 40%.**Answer** _____ students

- 3. Jacket Sale.** A jacket costs \$64. It is **25% off**. How much is taken off? What is the new price?

Step 1 Find 25% of \$64.

Step 2 Subtract that from \$64.

a. Discount: _____

b. Sale price: _____

Show your work.

- 4. Soccer Team.** A soccer team has 40 players. 15% of the players are new this year. How many players are new?

Step 1 Find 10% of 40, then 5% (half of 10%).

Step 2 Add them to get 15%.

Answer _____ players

- 5. Compare.** Which is greater: 30% of 90, or 90% of 30? Show both amounts. Explain what you notice.

Frame 30% of 90 = ____ and 90% of 30 = ____ . I notice ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Find a sale sign or a tip suggestion. Work out the exact amount with someone at home.

LESSON 4-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Compare Percents**

Maryland Standard 6.RP.A.3c · Learning target – I can compare two records fairly by converting each one to a percent.

REMEMBER

- Compare records fairly: write each as a **percent**.
- Scale to out of 100, or divide part ÷ whole.
- More hits is **not** always a better rate.

WORKED EXAMPLE

Mia made 14 of 20 shots. Leo made 18 of 25. Who has the better rate?

Step 1 Mia: $14/20 = 70/100 = 70\%$. Leo: $18/25 = 72/100 = 72\%$.**Step 2** Compare: $72\% > 70\%$.**Answer** Leo has the better rate.**WORD BANK****record** – part out of a whole, like 9 out of 20**percent** – out of 100**compare** – decide which is greater**better rate** – the greater percent**PRACTICE***show your work · label units*

1. Write each record as a percent.

Step 1 Write the record as a fraction.**Step 2** Make the bottom 100, or divide top by bottom.

- a. 21 out of 30 = ____ % b. 13 out of 20 = ____ % c. 9 out of 12 = ____ %

- 2.
- Softball Hits.**
- Complete the table. Circle the player with the best record.

Step 1 Divide hits by at bats for each player.**Step 2** Write each as a percent. Circle the greatest.

Player	Hits	At bats	Percent
Ava	9	20	
Ben	12	25	
Cruz	22	50	

- 3.
- Spelling Tests.**
- Rosa got 34 of 40 words
- right**
- . Kai got 42 of 50 words right. Write each as a percent. Who did better?

Step 1 Rosa: make 40 into 100. Kai: make 50 into 100.**Step 2** Compare the two percents.**Percents and who did better** _____

4. Select
- all**
- the records that are greater than 70%.

- ☐ **A** 15 out of 20 ☐ **B** 17 out of 25 ☐ **C** 36 out of 50 ☐ **D** 7 out of 10

5. **Find the Error.** Deja says $18 \text{ out of } 24 = 24 \text{ out of } 30$. Is she right? Explain.

Frame *Deja is ____ because $18/24 = \text{____}\%$ and $24/30 = \text{____}\%$.*

TEST PRACTICE

choose the best answer

6. Which record has the greatest percent?

A 13 out of 20 **B** 16 out of 25 **C** 33 out of 50 **D** 6 out of 10

7. Team A won 12 of 16 games. Team B won 15 of 25 games. Which is true?

Step 1 Find Team A's percent: $12 \div 16$.

Step 2 Find Team B's percent: $15 \div 25$.

Step 3 Pick the true sentence.

- A** B did better. It won more games. **B** A won 75%, B won 60%. A did better.
C Both teams did equally well. **D** A won 12%, B won 15%.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Compare two game records at home by changing both to percents.